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MZ 505
Inventory Systems

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Session 18: Risk Pooling I

Some slides adapted from Cachon and Terwisch (2013)

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Risk pooling in inventory management

- Risk pooling refers to a **strategy** that reduces the impact of demand variability on inventory investment by **consolidating inventory across multiple locations or products**.
  - The main idea is to "pool" resources, so that variations in demand across different areas or products can balance each other out, leading to more stable overall demand and, ultimately, lower inventory levels needed to meet customer needs.
- Types of risk pooling include: Product pooling, location pooling, capacity pooling, lead time pooling (consolidated distribution and delayed differentiation).

Product Pooling

Example: Product pooling

- O'Neill sells two Hammer 3/2 wetsuits that are identical except for the logo silk screened on the chest.

Surf Hammer 3/2 logo

Dive Hammer 3/2 logo



Performance without pooling

- For each product (surf hammer and dive hammer):
  - Demand is normally distributed with μ = 3192 and σ = 1181
  - Selling price = $190, unit cost = $110, salvage value = $90
- Two identical newsvendor type of problems:
  - $CR = \frac{C_u}{C_u + C_o} = \frac{80}{80+20} = 0.8$
  - $Q^* = \mu + z(0.8) \cdot \sigma = 3192 + 0.85 \cdot 1181 = 4196$
  - $E[\text{profit}] = pE[\text{sales}] + sE[\text{left}] - cQ = \$222{,}280$
- $\text{Total profit} = 2 \cdot 222{,}280 = \$444{,}560$

Example: Product pooling

- O'Neill sells two Hammer 3/2 wetsuits that are identical except for the logo silk screened on the chest.

Surf Hammer 3/2 logo

Dive Hammer 3/2 logo



- Instead of having two Hammer 3/2 suits, O'Neill could consolidate its product line into a single Hammer 3/2 suit, i.e., a universal design, which we will call the "Universal Hammer".

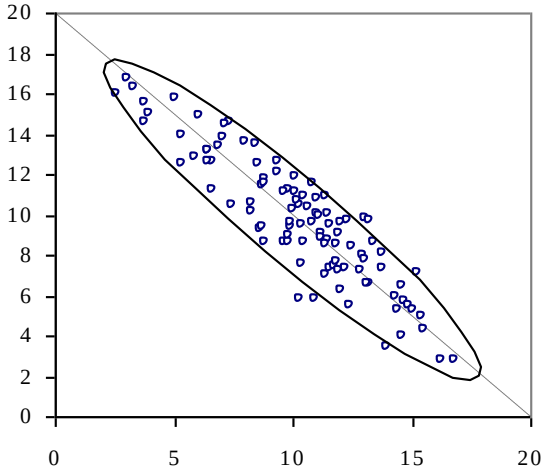
Performance with pooling

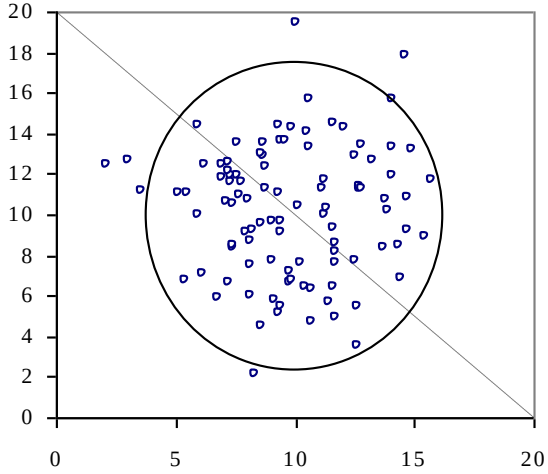
- For each product (surf hammer and dive hammer):
  - Demand is normally distributed with μ = 3192 and σ = 1181
  - Selling price = $190, unit cost = $110, salvage value = $90
- For the universal design:
  - Demand is normally distributed with
    - $\text{Mean} = 3{,}192 \cdot 2 = 6384$
    - $\text{Stdev} = 1181 \cdot \sqrt{2} = 1670$
  - $\text{CR} = \frac{C_u}{C_u + C_o} = \frac{80}{80+20} = 0.8$
  - $Q^* = \mu_u + z(0.8) \cdot \sigma_u = 6384 + 0.85 \cdot 1670 = 7804$
  - $\text{E[profit]} = \text{pE[sales]} + \text{sE[left]} - \text{cQ} = \$463{,}962$

4.4% profit increase with 7% decrease in inventory

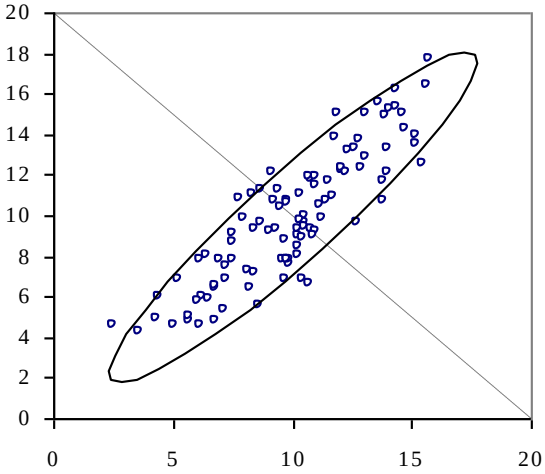
Product Pooling: Correlation

Correlation: quantitative measure of the strength and direction of the relationship between two random variables.





Random demand for two products (x-axis is product 1, y-axis is product 2). In scenario 1 (upper left graph) the correlation coefficient is -0.9, in scenario 2 (upper right graph) the correlation is 0 and in scenario 3 (the lower graph) the correlation is 0.9. In all scenarios demand is Normally distributed for each product with mean 10 and standard deviation 3.



Covariance and Correlation

- The covariance of two random variables X and Y, denoted by cov(X, Y) is defined by

$$\mathrm{cov}(X, Y) = E[\,(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

- The correlation coefficient ρ(X, Y) of two random variables X and Y that have nonzero variances is defined by

$$\rho(X, Y) = \frac{\mathrm{cov}(X,Y)}{\sqrt{\mathrm{var}(X)\mathrm{var}(Y)}} \quad \rightarrow \quad \rho \in [-1, 1]$$

- The variance of the sum of two random variables X and Y is

$$\mathrm{var}(X + Y) = \mathrm{var}(X) + \mathrm{var}(Y) + 2\mathrm{cov}(X, Y)$$

- More generally, for n random variables X_1, X_2,…, X_n:

$$\mathrm{var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{var}(X_i) + \sum_{\{(i,j)|i\neq j\}} \mathrm{cov}(X_i, X_j)$$

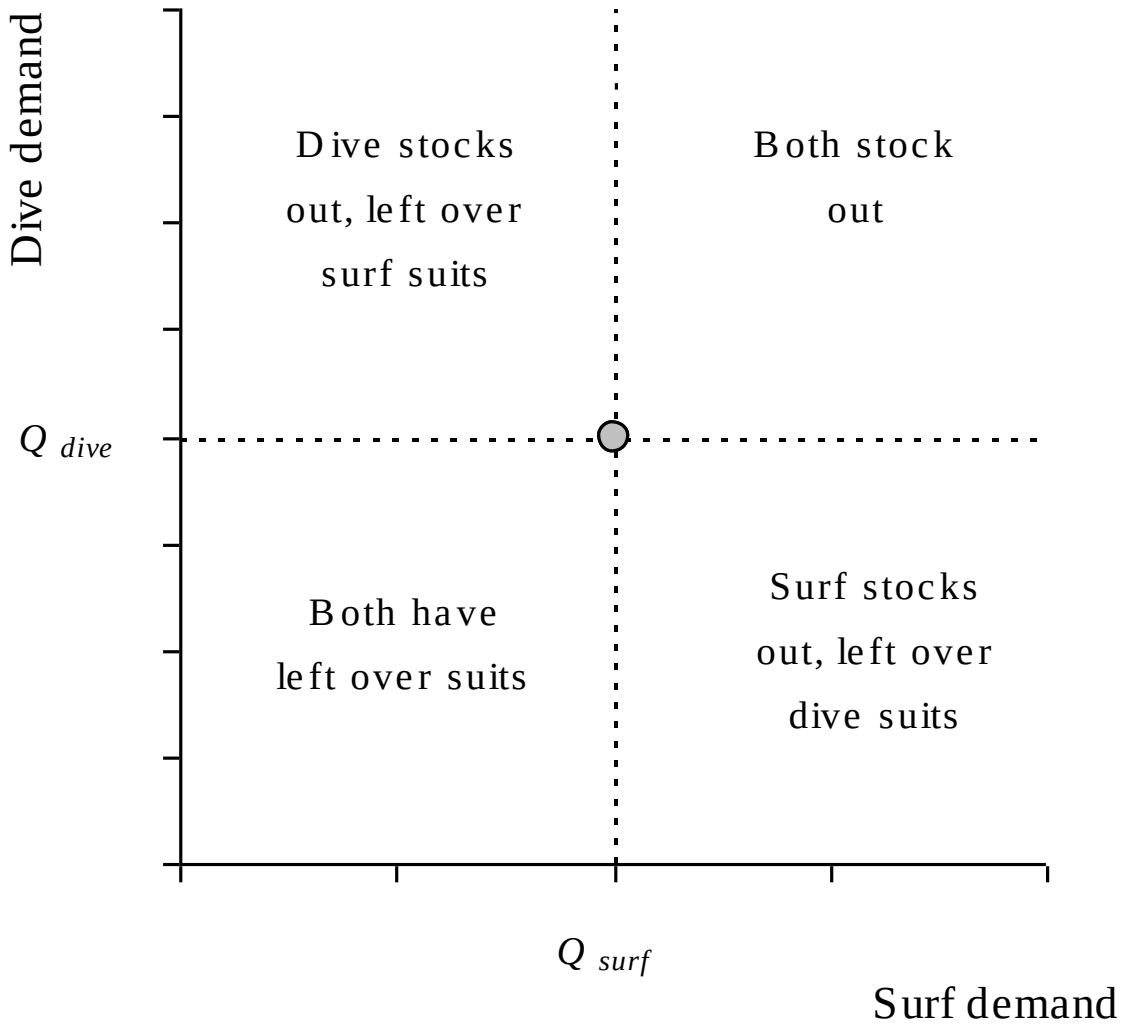
Pooling two identically distributed demands

- Consider two random variables X and Y that are identically distributed with mean μ, standard deviation σ, and correlation coefficient ρ.

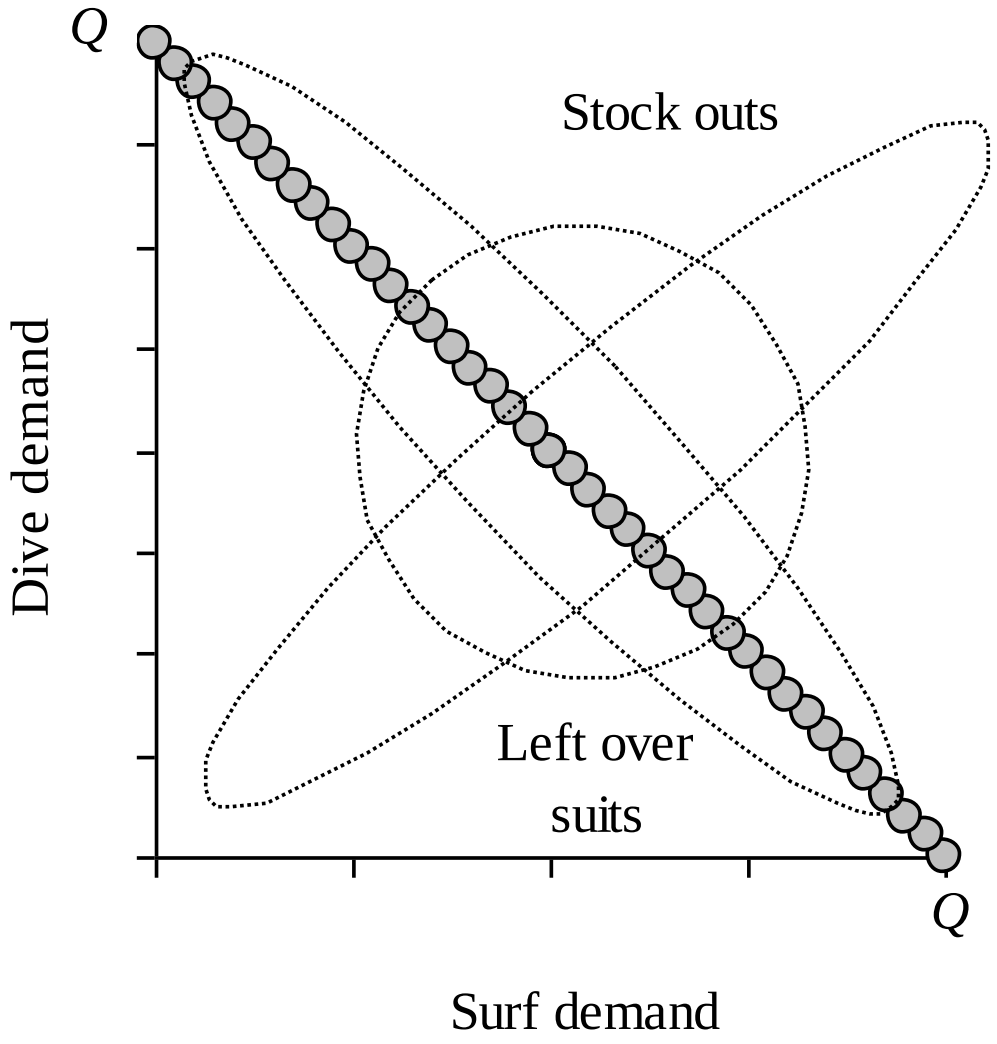
- Their sum (X + Y) will have
  - $E[X + Y] = E[X] + E[Y] = 2 \cdot \mu$
  - $\sigma(X + Y) = \sqrt{2 \cdot (1 + \rho)} \cdot \sigma$
  - Coefficient of variation:

$$\frac{\sqrt{2(1+\rho)} \cdot \sigma}{2 \cdot \mu} = \sqrt{0.5 \cdot (1 + \rho)} \cdot \frac{\sigma}{\mu}$$

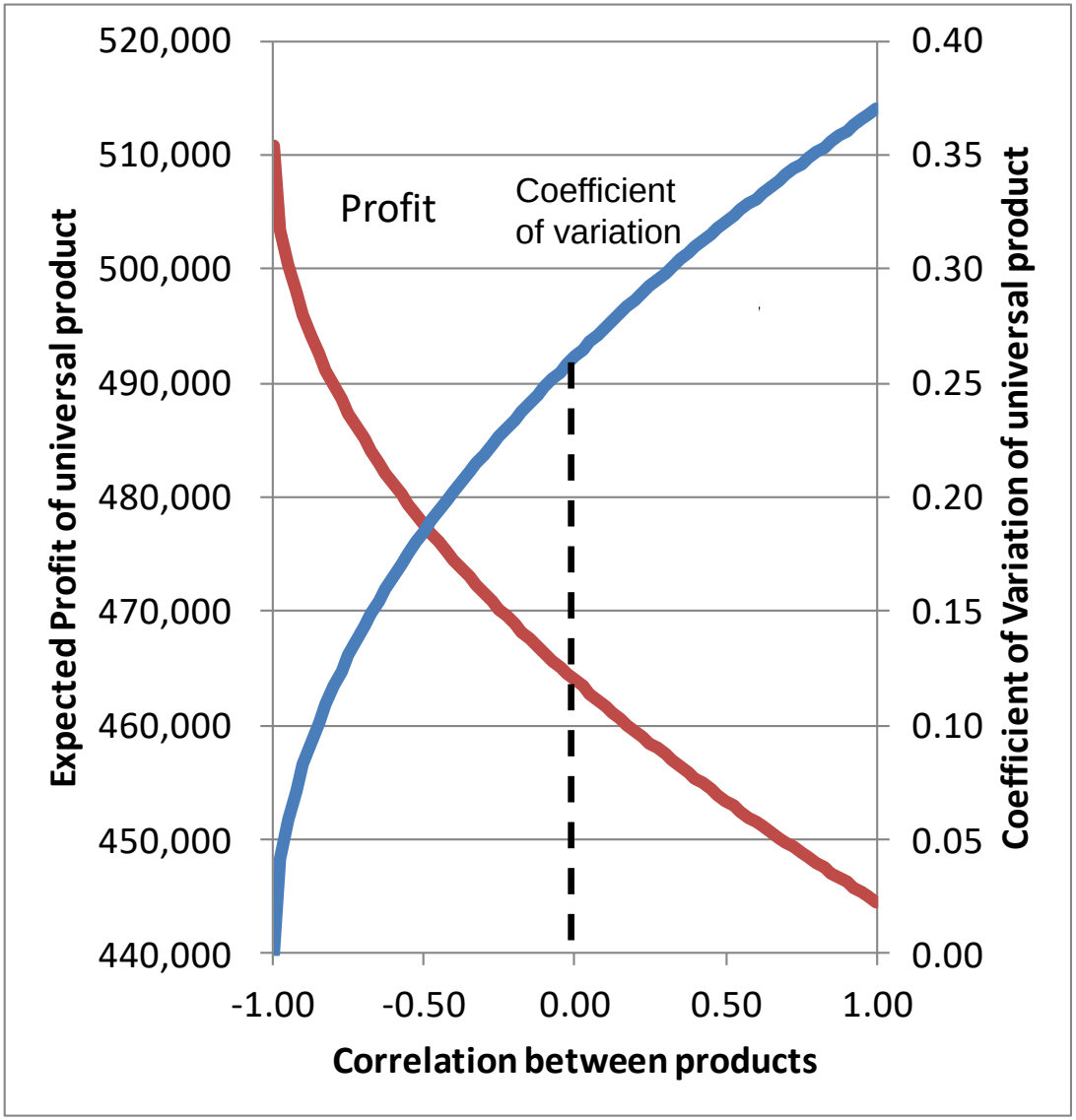
O'Neil example: No pooling



O'Neil example: Product pooling



O'Neil example: Profit with pooling vs. correlation

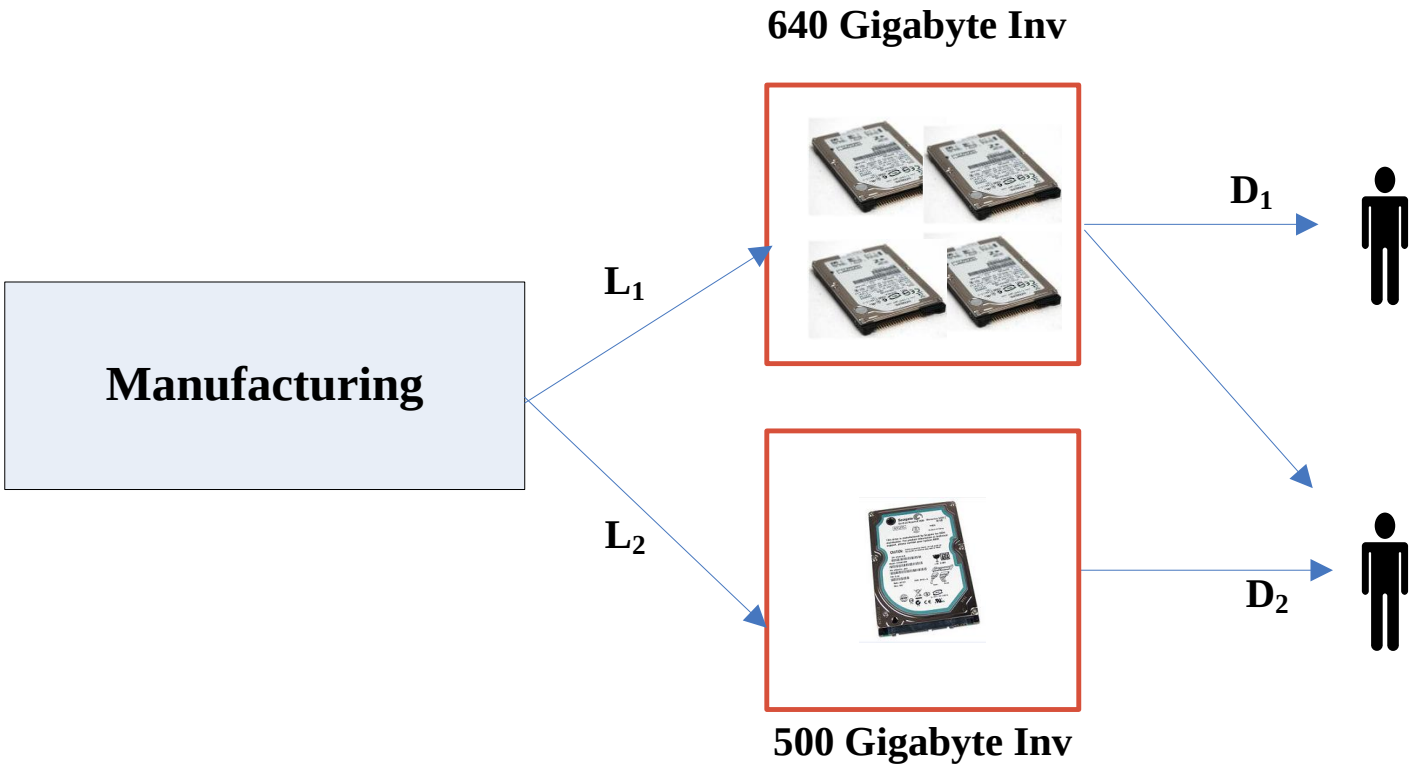


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Related concepts: Product substitution

- Substitution: use of one product to satisfy the demand for another product.
- Manufacturer-driven substitution
  - Decision of product substitution is made by the manufacturer.
  - Decision based on its own inventory position, market forecast and other related information.
  - Often occurs in multi-product manufacturing system.
- Customer-driven substitution
  - If a product type is unavailable, the customer might purchase a substitute product or might not purchase at all (lost sales).
  - Substitution causes the demand for the available products to increase, affecting their optimal inventory levels.

Example: Manufacturer-driven substitution

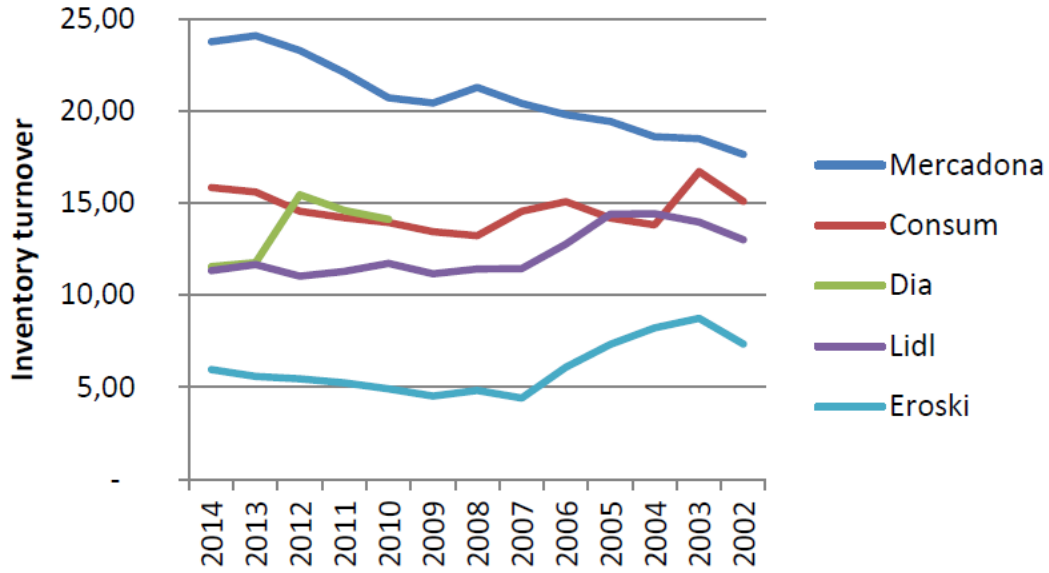


Product pooling: Application



vs.





On the "dark side of innovation"

- Companies create problems for customers and employees when product innovation goes unmanaged.
  - Eventually, excessive operational complexity hurts the bottom line.
  - Complexity: several systems; data duplication; multiple layers of decision-making; inconsistent service.

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- Leader in product innovation; in early 2000s, company's revenue plunged 40%, profits were wiped out, and market capitalization fell significantly.
- By 2016: the company moved from 6 to 1 BU; standardized systems around 3 areas (idea-to-market, market-to-order, order-to-cash); EBITDA > 10%; share price doubled since 2011.

(1) Instead of variety, focus on product integration (new features, cross-selling, bundling), and **(2)** Make sure your product developers work closely with customer-facing and operational employees.

Source: Mocker & Ross, "*The problem with product proliferation*" (Harvard Business Review, 2017)

Product pooling: Summary

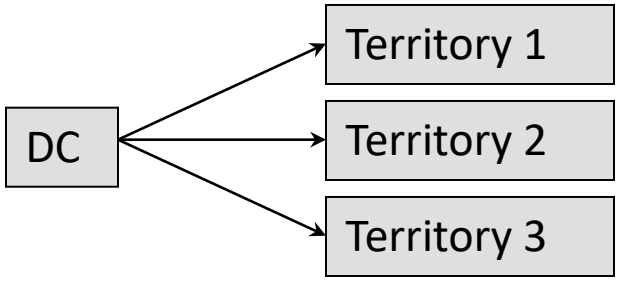
- The pooling benefits of serving the same aggregate demand with less product variety.
  - A key force behind the “SKU rationalization”.
- Advantages:
  - Can achieve the same service level with reduced inventory, or increase service level with the same inventory, or **both**.
- Disadvantages:
  - May limit product functionality; may increase production cost; brand/price segmentation issues.
  - Reducing variety can mean walking away from new revenue in the short term.

Location Pooling

Example: Location pooling at Medtronic

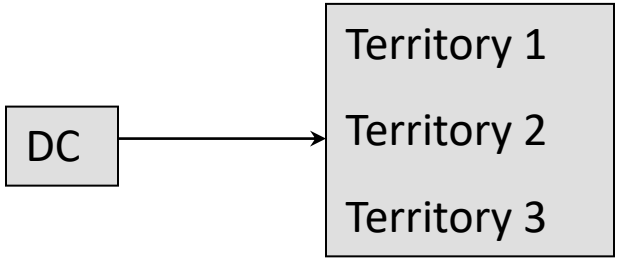
Current operations:

- Each sales representative have their own inventory to serve demand in their own territory (e.g., 3 territories, 3 stockpiles of inventory).
- Lead time is 1 day from Mounds View DC.



Location pooling strategy:

- A single location stores inventory used by several sales reps from adjacent locations.
- Sales reps no longer hold their own inventory, they must pull inventory from the pooled location (e.g., 3 pooled territories, 1 stockpile of inventory).
- Inventory is replenished at the pooled location as depleted by demand.
- Lead time to pooled location is still 1 day from Mounds View DC.



Inventory performance with location pooling

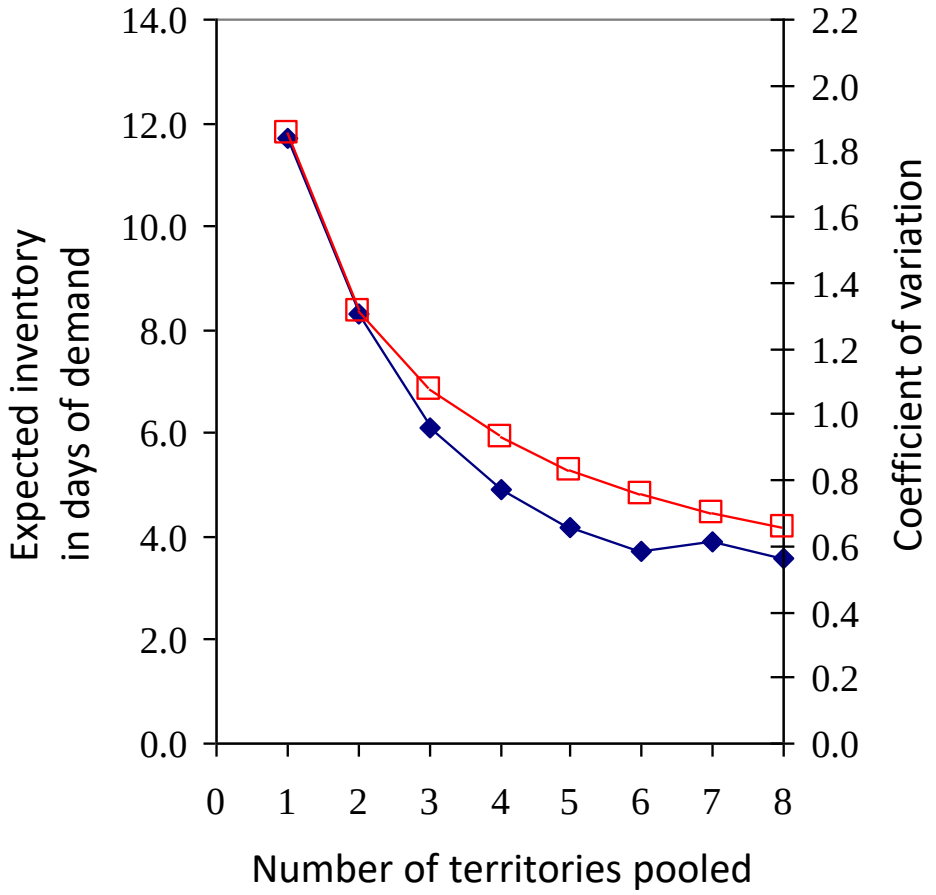
- Suppose each territory's expected daily demand is 0.29, the required in-stock probability is 99.9% and the lead time is 1 day with individual territories or pooled territories.

Number of territories pooled	Pooled territory's expected demand per day (a)	S	Expected inventory: On-hand units (b)	Expected inventory: days-of-demand (b/a)	Pipeline inventory: units (c)	Pipeline inventory: days-of-demand (c/a)
1	0.29	4	3.4	11.7	0.29	1.0
2	0.58	6	4.8	8.3	0.58	1.0
3	0.87	7	5.3	6.1	0.87	1.0
4	1.16	8	5.7	4.9	1.16	1.0
5	1.45	9	6.1	4.2	1.45	1.0
6	1.74	10	6.5	3.7	1.74	1.0
7	2.03	12	7.9	3.9	2.03	1.0
8	2.32	13	8.4	3.6	2.32	1.0

- Pooling 3 territories reduces expected inventory from 11.7 days-of-demand down to 6.1.
- But pooling has no impact on pipeline inventory.

Impact of location pooling on demand variability

- Location pooling reduces demand uncertainty as measured by the coefficient of variation.
- Subsequent reduction in inventory that is needed to achieve a target service level.
- But there are decreasing marginal returns to risk pooling.
  - Most of the benefit can be captured by pooling only a few territories.

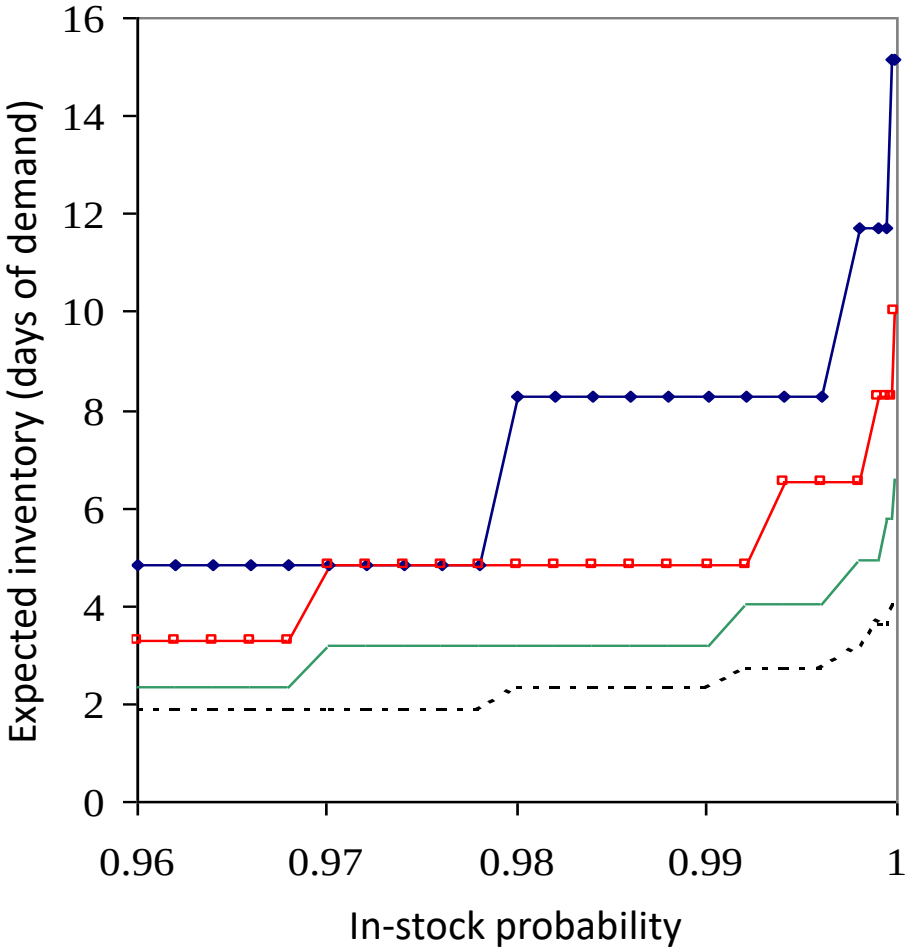


Graph: Expected inventory (diamonds) and coefficient of variation (squares) vs. # of pooled territories (Demand: Poisson w/ mean 0.29 ; Target in-stock probability 99% ; LT = 1 day)

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Location pooling: Inventory – Service tradeoff

- Location pooling shifts the inventory-service tradeoff curve down and to the right.
- For a single product, location pooling can be used to decrease inventory while holding service constant, or increase service while holding inventory constant, or a combination of inventory reduction and service increase.
- Location pooling can also be used to broaden the product line.



Inventory-service tradeoff curve for different levels of location pooling (1, 2, 4, and 8 pooled territories). (Demand: Poisson w/ mean 0.29 ; LT = 1 day)

Example: Location pooling with Normal demand

- Consider n regions, with demand in each region normally distributed with the following characteristics:
  - μ_i: Mean weekly demand in region i=1,…,n
  - σ_i: Standard deviation of weekly demand in region i=1,…,n
  - ρ_{ij}: Correlation of weekly demand for regions i, j (i≠j)
- There are two ways to serve demand in the n regions:
  - **Decentralized case**; i.e., have local inventories in each region.
  - **Centralized case**; i.e., aggregate inventories into one centralized facility.
- Assumption: continuous review of inventory with lead time L and targeted cycle service level CSL.

Impact of location pooling on safety stock

- Total safety stock for decentralized case:

$$SS^D = \sum_{i=1}^{n} SS_i = \Phi^{-1}(CSL) \cdot \sqrt{L} \cdot \sum_{i=1}^{n} \sigma_i$$

- For centralized case: the aggregate demand is normally distributed with a mean of D^C and standard deviation of σ^C

$$\mu^C = \sum_{i=1}^{n} \mu_i \qquad \sigma^C = \sqrt{\sum_{i=1}^{n} \sigma_i{}^2 + \sum_{i \neq j} \rho_{ij} \sigma_i \sigma_j}$$

- Safety stock for centralized case:

$$SS^C = \Phi^{-1}(CSL) \cdot \sqrt{L} \cdot \sigma^C$$

- Inventory reduction:

$$\Phi^{-1}(CSL) \cdot \sqrt{L} \cdot \left(\sum_{i=1}^{n} \sigma_i - \sigma^C\right)$$

Impact of location pooling on safety stock (cont.)

- If demands in all regions are <u>independent</u> ($\rho_{ij} = 0$) and <u>identically distributed</u> with mean D and standard deviation σ, then

$$\mu^C = n \cdot \mu \qquad \sigma^C = \sigma\sqrt{n}$$

$$SS^D = \Phi^{-1}(CSL) \cdot \sqrt{L} \cdot n \cdot \sigma$$

$$SS^C = \Phi^{-1}(CSL) \cdot \sqrt{L} \cdot \sqrt{n} \cdot \sigma$$

$$\frac{SS^C}{SS^D} = \frac{1}{\sqrt{n}}$$

- If number of independent stocking locations decreases by n, the expected level of safety inventory will be reduced by square root of n (**square root law**).

Example: Impact of correlation on value of pooling

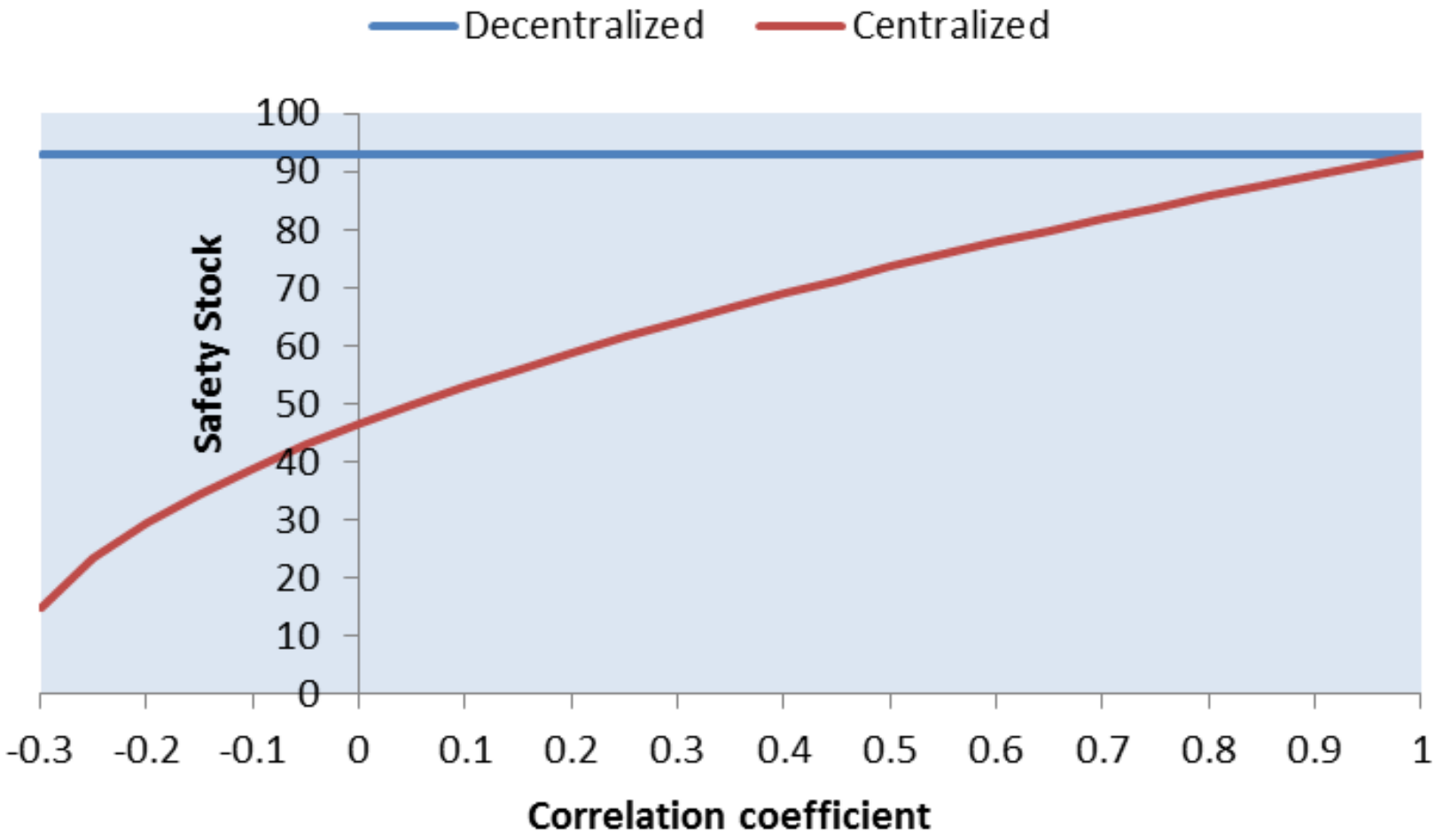
- n=4; D =50/week; σ = 10; L = 2 weeks; CSL = 0.95
- Compare needed SS in the two options as the correlation coefficient ρ varies between -1 and 1.
- If ρ = 0 (i.e., uncorrelated demand)
  - Total safety stock for decentralized case:

$$SS^D = \Phi^{-1}(CSL) \cdot \sqrt{L} \cdot \sum_{i=1}^{n} \sigma = \Phi^{-1}(0.95) \cdot \sqrt{2} \cdot 4 \cdot 10 = 93$$

  - Safety stock for centralized case (for ρ=0):

$$SS^C = \Phi^{-1}(CSL) \cdot \sqrt{L} \cdot \sqrt{n} \cdot \sigma = \Phi^{-1}(0.95) \cdot \sqrt{2} \cdot \sqrt{4} \cdot 10 = 46.5$$

Example: Impact of correlation on value of pooling



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Related concepts: Information centralization

- **Virtual pooling**
- Information system that allows access to current inventory records in all warehouses from each warehouse.
- Most orders are filled from closest warehouse.
  - In case of a stockout, another warehouse can fill the order.
- Better responsiveness, higher product availability, and reduced safety inventory… but tradeoff with transportation cost.
- Examples: Gap, Wal-Mart

Location pooling: Summary

- The pooling benefits of serving the same aggregate demand from fewer locations.
- Demand variability is reduced because as demand is aggregated across different locations, it becomes more likely that high demand from one customer (location) will be offset by low demand from another.
- Issues to consider:
  - Service level improvement vs. response time to customer order.
  - Overhead cost savings and inbound economies of scale vs. increased transportation costs to customer.
- Location pooling is less effective as demands become more positively correlated.

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Location pooling: Application



Graph: Location of Amazon Fulfillment Warehouses across the USA (2015)

Location pooling: Application

Heavy-tailed distributions are used to model uncertainty when extreme events, i.e., very large or very small outcomes, are relatively likely, e.g., much more likely compared to what a normal distribution would have predicted. Arguably, markets for fashion, technology (e.g., tablets and smartphones) and creative goods (e.g., movies and books), where there is significant uncertainty with respect to the number of sales of a new product, fall into this category. Chevalier and Goolsbee (2003) estimate that the empirical distribution of book demand at Amazon has a power-law tail with exponent $\alpha = 1.2$, whereas Gaffeo et al. (2008) report that the tail exponents for book sales in Italy lie in the interval $\alpha \in (1, 1.4)$, and conclude that "... the values of α are always significantly different from 2 and, therefore, the estimated degree of uncertainty in the book publishing market is too high to be compatible with a Gaussian distribution."

Bimpikis & Markakis, 2016 ,"Inventory pooling under heavy-tailed demand", Management Science. 62(6) (very nice literature review, multi-location identical NVs)

$$\text{Prob}(D > x) \cong \frac{1}{x^a} \rightarrow \frac{SS^C}{SS^D} = n^{\frac{1-a}{a}}$$

Readings

- Chapter 15 (sections 15.1 and 15.2) from Cachon & Terwiesch, 3rd ed.

Thank you!